

Granular Credit Risk

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- What is the impact of granular credit risk on banks and the economy?
- Granular Hypothesis (Gabaix 2011) - shocks to large agents matter
- Analysis of the "granularity" of the banking sector in Norway
- Loan share distribution is fat-tailed
- GIV allows to show that idiosyncratic risk does not disappear at the aggregate level
- Several implications:
 - shocks spill over from affected banks to firms
 - this reduces investment
 - raises probability of default among non-granular borrowers

Dataset

- Unit of observation is an **individual loan**, with annual frequency
- For each **loan** they observe:
 - firm-bank identifiers
 - flow of interest paid during the year
 - end-of-year stock of debt
- Dataset covers all limited liability companies, 2003-2015 period, roughly 90% of private sector employment
- Credit rating agency data from Bisnode - age, location, industry, credit score, income statements - for all Norwegians firms for 1999-2019
- Bank data from ORBOF - annual balance sheet and income statement covering all Norwegian banks over 1987-2019

Key variables (I/II)

Return on a loan (RoL)

$$RoL_t = \frac{R_t}{0.5D_{t-1} + 0.5D_t}$$

- RoL_t - ex-post collected interest rate
- R_t - sum of interest actually received in year t
- D_t - end-of-year stock of outstanding debt

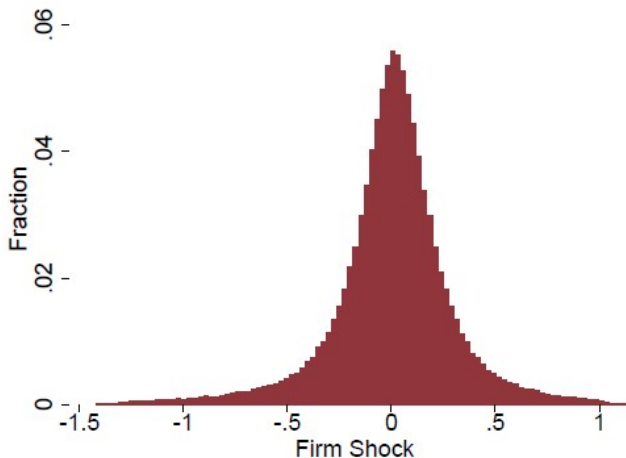
Key variables (II/II)

Idiosyncratic Firm Shocks. To estimate firm-level shocks, they regress (log) value-added:

$$\ln VA_{j,t} = \alpha_j + \theta_{g(j),t} + \beta_1 \ln K_{j,t} + \beta_2 \ln W_{j,t} + X'_{j,t} \lambda + \varepsilon_{j,t} \quad (1)$$

- $VA_{j,t}$ - firm value added (total sales minus intermediate inputs like materials, energy)
- $K_{j,t}$ - book capital
- $W_{j,t}$ - wage bill
- $X'_{j,t}$ - firm leverage, liquidity, credit rating and a quadratic in firm age
- α_j - firm fixed effect absorbs time-invariant characteristics
- $\theta_{g(j),t}$ - year-by-industry-by-county common characteristics
- $\varepsilon_{j,t}$ - realized performance shock that was unlikely to be forecastable by banks (mismanagement, legal issues, accidents)

Figure 1: Distribution of Idiosyncratic Firm Shocks



Notes: This graph plots the pooled distribution of idiosyncratic firm shocks estimated from equation (1).

First empirical fact: loan portfolios are granular. They calculate excess loan portfolio Herfindahls (eHHI), pooled across banks and years. The eHHI is defined as:

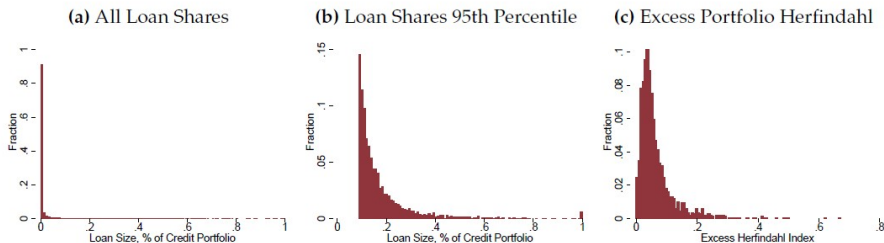
$$eHHI_{i,t} = \sqrt{\sum_j s_{i,j,t}^2 - \frac{1}{N_{i,t}}} \quad (2)$$

- $s_{i,j,t}$ - loan share of firm j in bank i 's portfolio in year t
- $N_{i,t}$ - number of loans in the respective portfolio

The higher the eHHI, the higher the concentration. Based on the Figure 2, they suspect that loan-level disturbances do not average out in the aggregate. 80% of all credit is concentrated in 20% of the loans.

Granularity of the Distribution of Loan Shares

Figure 2: Granularity in the Distribution of Bank Loan Portfolios



Notes: This graph presents the distribution of bank loan shares and of excess portfolio Herfindahls. Panel (a) plots the full distribution of loan shares. Panel (b) zooms in on the 95th percentile. The share of each loan is computed as the ratio of a singular loan's amount to total corporate loans of a given bank in a given year. The panels plot the pooled shares for all banks and years. The Pareto rate of the 95th percentile is 1.82. Panel (c) plots the distribution of excess loan portfolio Herfindahls (eHHI), as defined in main text.

Loan Outcomes

Impact of idiosyncratic firms shocks on loan-level returns. For firm j , in the set of all firms that borrow from bank i in year t , $\mathcal{P}(i, t)$:

$$R_{i,j,t} = \beta \varepsilon_{j,t} + \theta_{g(i,j,t),i,t} + \alpha_j + \nu_{i,j,t} \quad (3)$$

- α_j - firm fixed effect
- $\theta_{g(i,j,t),i,t}$ - in most conservative specification, it includes the full interaction of bank, year, firm, industry, county, loan type fixed effects
- $\varepsilon_{j,t}$ - performance shock for firm j at time t

Loan Outcomes - results

Table 1: Loan Outcomes

	(1)	(2)	(3)	(4)
	Depend. Variable: Return on Loan (RoL)			
Firm Shock (std.)	0.334 (0.015)	0.335 (0.017)	0.361 (0.019)	0.336 (0.017)
Bank x Industry x Year FE	-	✓	-	-
Bank x Industry x Year x Loan-type x County FE	-	-	✓	✓
Firm FE	-	-	-	✓
Observations	333289	317186	292825	282002
R ²	0.001	0.114	0.167	0.528
E(RoL)	9.012%	9.029%	9.098%	9.076%
SD(RoL)	8.921%	8.928%	8.923%	8.687%

Notes: This table reports results from the regression of loan-level returns on idiosyncratic firm shocks. The exact specification is described by equation (3). Firm shocks are normalized by their standard deviation. Loan types include regular and credit-line loans. Counties are 19 administrative areas (*fylke*) in Norway. Industries are 99 2-digit sectors. Standard errors (in parentheses) are double clustered at the firm and year levels and computed by bootstrapping with 5,000 replications. The last two rows report the unconditional sample mean and standard deviation of the dependent variable.

A one std. increase in the firm shock measure affects loan returns by 33.5 basis points, about 4% of the dependent variable's standard deviation.

Bank level outcomes

- Aggregate the firm shock measure

$$\bar{\epsilon}_{i,t} = \sum_{j \in \mathcal{P}(i,t)} s_{i,j,t} \epsilon_{j,t} \quad (4)$$

- Dependent variable

$$R_{i,t}^b = \alpha_i + \alpha_t + \beta \bar{\epsilon}_{i,t} + \omega'_{i,t} \gamma + \nu_{i,t} \quad (5)$$

- $R_{i,t}^b$ - bank-level return on all corporate loans (RoA)
- $\omega_{i,t}$ - vector of observable bank-level controls
- α_i and α_t - bank and time fixed effects
- ν - residual defined to be orthogonal to the control vector

Identification

Problem: At the loan level, they exploited within-bank-year variation to control for credit supply-side factors. Now, endogeneity:

$$\epsilon_{j,t} = \eta'_{i,t} \delta_i + u_{i,j,t}, \quad \forall j \in \mathcal{P}(i, t) \quad (6)$$

- $\eta_{i,t}$ - vector of bank characteristics
- $u_{i,j,t}$ - residual of firm j 's shock, orthogonal to bank i 's characteristics
- When η and ν are correlated, η also affects bank returns and is inside the error term ν , $\text{Corr}(\bar{\epsilon}_{i,t}, \nu_{i,t}) \neq 0$

Firm shocks can be contaminated by uncaptured time-varying bank characteristics η'

Identification

Solution: Granular Instrumental variable. They construct an instrument for the "endogenous" covariate $\bar{\epsilon}_{i,t}$. For a loan portfolio of bank i in year t , subtract unweighted average of firm shocks from the loan-share weighted average of firm shocks.

$$Z_{i,t}^{\text{GIV}} = \sum_{j \in \mathcal{P}(i,t)} \left(s_{i,j,t} - \frac{1}{N_{i,t}} \right) \epsilon_{j,t} \quad (7)$$

$$\text{rewritten as: } Z_{i,t}^{\text{GIV}} = \sum_{j \in \mathcal{P}(i,t)} s_{i,j,t} \epsilon_{i,j,t}^* \quad (8)$$

$$\text{where: } \epsilon_{i,j,t}^* = \epsilon_{j,t} - \frac{1}{N_{i,t}} \sum_{j \in \mathcal{P}(i,t)} \epsilon_{j,t}, \quad \forall j \in \mathcal{P}(i,t) \quad (9)$$

$\epsilon_{i,j,t}^*$ is the shock of firm j after the subtraction of the average firm shock in bank's portfolio

Why does it work?

Substitute the decomposition:

$$\epsilon_{i,j,t}^* = (\eta'_{i,t}\delta_i + u_{i,j,t}) - \frac{1}{N_{i,t}} \sum_{j \in \mathcal{P}(i,t)} (\eta'_{i,t}\delta_i + u_{i,j,t}) \quad (10)$$

Now split the average:

$$\epsilon_{i,j,t}^* = \eta'_{i,t}\delta_i + u_{i,j,t} - \left[\eta'_{i,t}\delta_i + \frac{1}{N_{i,t}} \sum_{j \in \mathcal{P}(i,t)} u_{i,j,t} \right] \quad (11)$$

The bank-year factor cancels:

$$\eta'_{i,t}\delta_i - \eta'_{i,t}\delta_i = 0 \quad (12)$$

So we get:

$$\epsilon_{i,j,t}^* = u_{i,j,t} - \frac{1}{N_{i,t}} \sum_{j \in \mathcal{P}(i,t)} u_{i,j,t} \quad (13)$$

Exclusion Restriction

$$\mathbb{E} \left[\sum_{j \in \mathcal{P}(i,t)} \underbrace{s_{i,j,t}}_{\text{Weights}} \underbrace{\epsilon_{i,j,t}^*}_{\text{Firm Shocks}} \nu_{i,t} \right] = 0 \quad (14)$$

They argue that this is satisfied because firm shocks $\epsilon_{i,j,t}^*$ are as-good-as randomly assigned.

- they show robustness in the appendix.

Instrument Relevancy

$$\text{Corr}(Z^{\text{GIV}}, \bar{\epsilon}) = \sqrt{\frac{\text{eHHI}}{\frac{\sigma_{\eta}^2}{\sigma_u^2} + \text{HHI}}} \quad (15)$$

- Banking sector features a fat tail - very concentrated
- Aggregate excess loan share Herfindahl for the whole banking sector is persistently positive, never vanishes to zero
- Pearson correlation between the instrument Z^{GIV} and endogenous covariate $\bar{\epsilon}_{i,t}$ is equal to 0.863 - very high.
- Their main bank-level specification is then a simple two-stage least squares regression with Z^{GIV} as instrument for $\bar{\epsilon}_{i,t}$

Bank Outcomes - results

Table 2: Bank Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent Variable: Bank Return on Loans (RoA)								
	OLS		Instrumented with GIV					
	Pooled	Pooled	Pooled	Positive	Negative	Pooled	Positive	Negative
Weighted Firm Shock	0.129 (0.029)	0.136 (0.027)	0.116 (0.031)	0.016 (0.094)	0.194 (0.074)	0.117 (0.030)	0.056 (0.087)	0.176 (0.072)
F-stat			1429.683	138.772	396.907	1137.722	150.136	263.982
Bank FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Bank Controls	-	✓	-	-	-	✓	✓	✓
Observations	1211	1211	1211	508	694	1211	508	694
R ²	0.752	0.770	0.599	0.646	0.569	0.627	0.683	0.592
E[RoA]	6.350%	6.350%	6.350%	6.460%	6.289%	6.350%	6.460%	6.289%
SD(RoA)	1.354	1.354	1.354	1.403	1.295	1.354	1.403	1.295

results show that a one standard deviation GIV-instrumented firm shock, on average, affects bank loan portfolio returns by 11-12 bps

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SD(RoA)	1.354	1.354	1.354	1.403	1.295	1.354	1.403	1.295

Asymmetric effects of granular firm risk - only negative shocks have a significant impact on bank returns.

Figure 3: Granular Credit Risk and Bank Outcomes

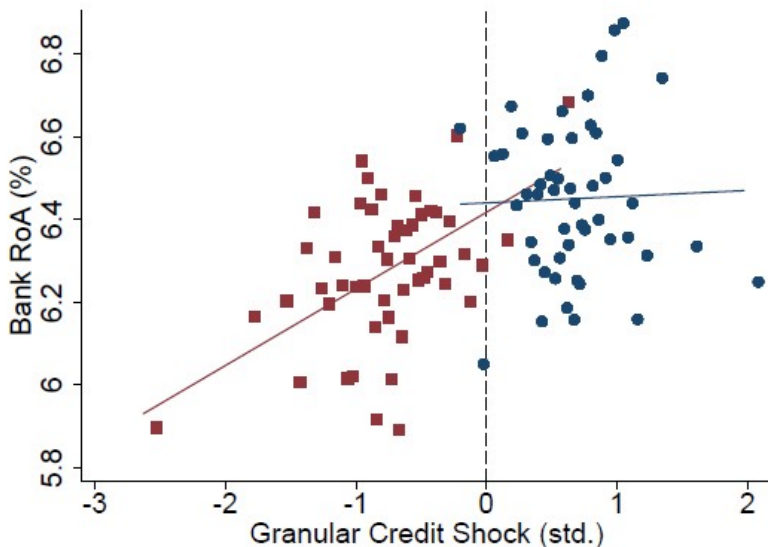


Table 3: Hedging Granular Credit Risk

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: Income from	Fees	Derivatives	Equity	Bonds	Dividends
	Pooled				
GIV-Instrumented Firm Shock (std.)	-0.000 (0.002)	0.001 (0.001)	0.016 (0.016)	0.003 (0.002)	0.002 (0.002)
Bank FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	1211	1210	1103	1198	1181
R ²	0.025	0.012	0.011	0.008	0.017
	Negative Shocks Only				
GIV-Instrumented Firm Shock (std.)	0.003 (0.003)	0.004 (0.002)	-0.000 (0.000)	0.006 (0.004)	0.003 (0.003)
Bank FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	697	679	632	692	684
R ²	0.049	0.043	0.273	0.044	0.032

If banks are hit with bad idiosyncratic shocks, returns could be compensated by different revenues/departments. Data cannot reject the hypothesis of little to no insurance against granular credit risk.

Table 4: Spillovers from Granular Credit Shocks: Loan-Level Supply

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: Δ Loans (std.)				
Δ Bank Shock (std.)	0.007 (0.013)	0.056 (0.037)	0.051 (0.038)	0.246 (0.115)	0.231 (0.112)
Non-Granular Firms (50%)	-	✓	✓	-	-
Non-Granular Firms (20%)	-	-	-	✓	✓
Year $\times$ Industry $\times$ County $\times$ Firm FE	✓	✓	✓	✓	✓
Bank FE	-	-	✓	-	✓
Instrumented by GIV	✓	✓	✓	✓	✓
Observations	15279	3479	3443	232	212

Overall, a one standard deviation negative granular credit shock reduces loan supply growth to non-granular borrowers by up to 5% (24%) of the dependent variable's standard deviation in the case of 50th (20th) percentile thresholds.

Table 6: Firm Outcomes from Granular Credit Shocks

	(1)	(2)	(3)	(4)	(5)
Dependent Variable: Δ Capital (std.)					
Δ Bank Shock (std.)	0.005 (0.003)	0.025 (0.009)	0.037 (0.011)	0.044 (0.025)	0.025 (0.059)
Non-Granular Firms (50%)	-	✓	✓	-	-
Non-Granular Firms (20%)	-	-	-	✓	✓
Industry x County x Year FE	✓	✓	✓	✓	✓
Firm FE	-	-	✓	-	✓
Instrumented by GIV	✓	✓	✓	✓	✓
Observations	157642	66648	55770	19608	13719

A one std. negative shock causes a decline in firms' fixed capital investment growth by roughly 2.5% - 4.4% - meaning that financial frictions can affect the real side of the economy.

Table 7: Firm Bankruptcy from Granular Credit Shocks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Dependent Variable: Prob. of Bankruptcy _t						Pr.(Ever Bankrupt)	
$\Delta \text{BankShock}_{t-1}$	-0.010 (0.010)	-0.010 (0.009)	-0.032 (0.010)	-0.033 (0.008)	-0.076 (0.046)	-0.090 (0.049)	-0.050 (0.009)	-0.095 (0.017)
Non-Granular Firms (50%)	-	-	✓	✓	-	-	✓	-
Non-Granular Firms (20%)	-	-	-	-	✓	✓	-	✓
Firm Controls	-	✓	-	✓	-	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Pseudo R ²	0.039	0.096	0.039	0.100	0.037	0.089	0.039	0.034
Observations	165000	165000	78511	78511	27828	27828	79965	28754

A one std. negative granular credit shock raises the probability of bankruptcy for non-granular firms by around 3% - 9%. Negative granular credit risk shocks are positively associated with the likelihood of corporate bankruptcy.